Sr.No. 8376

Bharatiya Vidya Bhavan's

SARDAR PATEL INSTITUTE OF TECHONOLOGY

(An Autonomous Institute Affiliated to University of Mumbai)

MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058

MID SEMESTER EXAMINATION

AY: 2025-26	Session: Odd MSE October 2025	Date: 08-10-2025
Sem: 1	Slot: 10:00am - 11:00am	C. Code: CS413
Exam Type: Written	Block:2D2 Course Section:1	Seat No: 2022600012

L_61_470_986_1_1505_48253

Name of the candidate. Piyanshu Gelani

Candidate's UID number:

2	0	2	2	6	0	0	0	1	2

Examination class room number:

E 2	0	2

Invigilator's signature with date: 08/10/2025

(To be filled in by the candidate)

EXAMINATION: BE B Tech MSE

(For example FY MCA/FY BTECH)

SEMESTER : Odd

I / II / III / IV / V / VI / VII / VIII

PROGRAM: CSE-AIML

DAY AND DATE: wednesday, 08/10/25

COURSE NAME: DL

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.

2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.

3) Do not write anything in the column provided for the marks to be assigned by the examiners.

4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.

5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.

6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.

7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.

8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**

9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.

10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.

11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.

12) The answer-books will be scrutinized before they are sent to examiners.

13) All answer-books supplied shall be returned whether written or blank.

14) Nothing shall be written on the question-paper.

15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.

16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.

17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Space For Marks	Question No.	START WRITING HERE (Begin answer for each question on a new page)

(Q1) a) $f(w) = \frac{1}{2}|xw - y|^2$ —(i)

Gradient update rule formula is as follows:

$$\boxed{w_{k+1} = w_k - \eta \cdot f'(w)} \text{ —(ii)}$$

Now, we need to find $f'(w)$.

$$f'(w) = \frac{\partial}{\partial w}(f(w))$$

$$= \frac{1}{2}\frac{\partial}{\partial w}|xw - y|^2$$

$$= \frac{1}{\cancel{2}}(\cancel{2}(xw-y)).(x)$$

$$\boxed{f'(w) = x(xw-y)}$$

Putting this in (ii),

$$\boxed{w_{k+1} = w_k - \eta(x^2 w - xy)}$$

is the derived gradient update rule.

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	Q1)B)	

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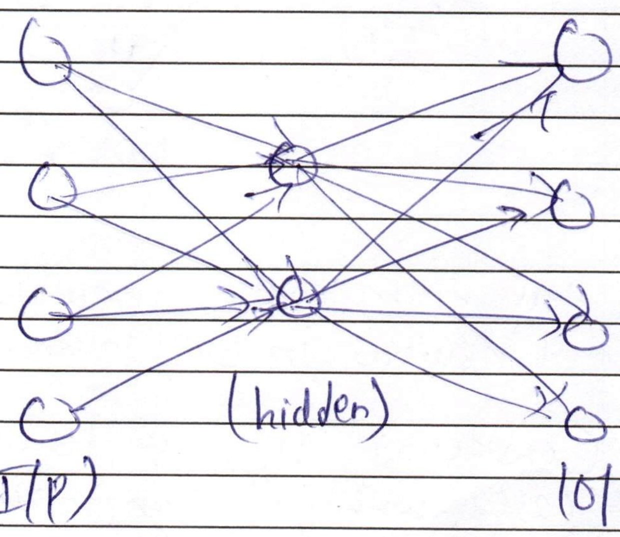
Space For Marks	Question No.	START WRITING HERE (Begin answer for each question on a new page)

Q2) b) Autoencoders are nothing but unsupervised neural networks used in image compression & expansion.

An undercomplete autoencoder ~~is~~ consists of hidden layer neurons lesser than input layer's.

eg.



Now in this simple ~~cut~~ undercomplete autoencoder, we can see number of neurons in hidden layer = 2 < i/p = 4.
2 neurons

Which means the hidden layer has to learn the patterns of 4 equivalent neurons.

This is basically feature extraction where the higher dimensional image is learned into lower dimensions while extracting information as much as possible. This exactly is used in Principal Component Analysis where top principal components are chosen.
(eigenvectors)

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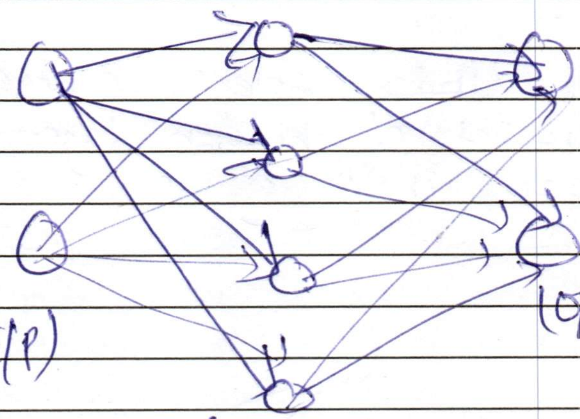
Q2) b) contd

An overautoencoder ~~is~~ consists of neurons greater than hidden layer.

eg.



Here there are 2 neurons in i/p layer & 4 neurons in o/p layer.

Consider if $n_{input} = 10$ & $n_{hidden} = 100$. The case with it can literally copy the image leads to lazy learning meaning it would not 'learn' the ~~...~~ deep patterns of input image.

without regularization, it will not be able to compress different patterns. Thus we use sparsity constraints and denoising techiniques which basically helps model learn the patterns

— Sparsity means we set a few neurons of hidden layer to zero, which forces the ~~other~~ neurons to learn 'edges' & other features.

— Denoising means we add noise to i/p image. ~~*~~

$x + \nabla x \longrightarrow$ i/p given to hidden layer. The hidden layer has to get the purified image 'x'. This is where universal approximation algorithm is also used.

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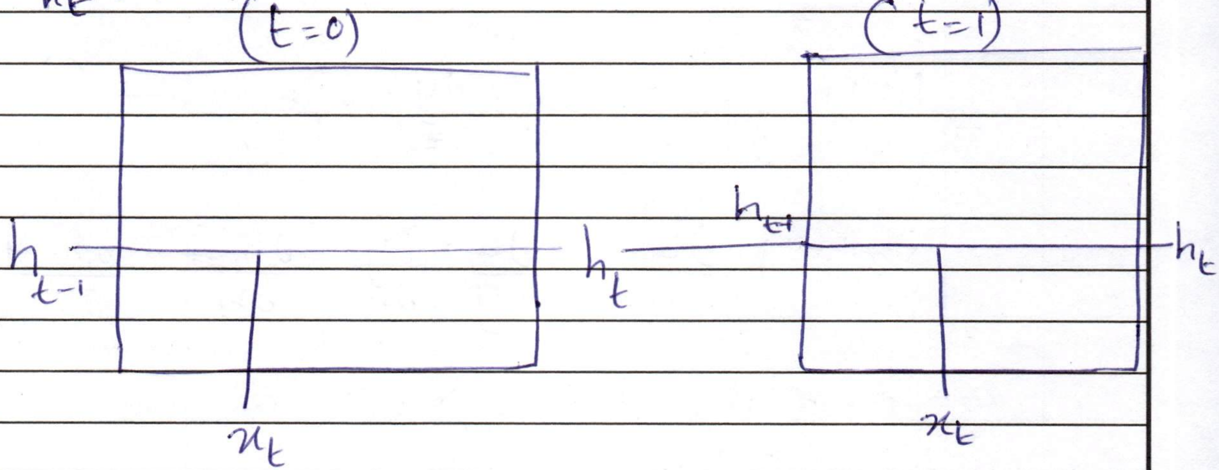
(Q3) a) RNN model is ~~a~~ recurrent neural network, meaning it takes h_{t-1} to its input for calculating h_t.



Now, let h_t be activation function.
∴ h_{t-1} will be activation function of previous timestamp of same layer.
∴ h_{t-k} will be activation function k timestamps before.

$$\boxed{h_t = \tanh(W_{aa}\cdot h_{t-1} + W_{ax}\cdot x_t + b_h)}$$

for simplicity we can write $a_t = h_t$.

$$\boxed{a_t = \tanh(W_{aa}\, a_{t-1} + W_{ax}\cdot x_t + b_a)}$$

& $\boxed{y = \text{softmax}(W_{ya}\cdot a_t + b_y)}$

Let o → actual o/p & y → predicted o/p.
∴ Loss = L.

Now, $L \rightarrow o(y) \rightarrow \underset{}{a_t} \, (h_t) \rightarrow h_{t-1}$

↳ $a_{t-1} \rightarrow h_{t-2}$

⋮

↳ $a_{t-k} \rightarrow h_{t-k}$

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Q3) a) Contd

To find : $\frac{\partial L}{\partial h_{t-k}} = \cancel{\frac{\partial L}{\partial y} \cdot \frac{\partial y}{\partial a_t} \cdot \frac{\partial a_t}{\partial h_{t-1}}}$

Now, $\frac{\partial L}{\partial h_{t-1}} = \frac{\partial L}{\partial y} \cdot \frac{\partial y}{\partial h_t} \cdot \frac{\partial h_t}{\partial h_{t-1}}$ —(i)

Similarly $\frac{\partial L}{\partial h_{t-2}} = \frac{\partial L}{\partial y} \cdot \frac{\partial y}{\partial h_t} \cdot \frac{\partial h_t}{\partial h_{t-1}} \cdot \frac{\partial h_{t-1}}{\partial h_{t-2}}$ —(ii)

this can be written as

$$\boxed{\frac{\partial L}{\partial h_{t-2}} = \frac{\partial L}{\partial h_{t-1}} \cdot \frac{\partial h_{t-1}}{\partial h_{t-2}}}$$

Generalizing it for $\frac{\partial L}{\partial h_{t-k}}$ we get,

$$\boxed{\frac{\partial L}{\partial h_{t-k}} = \frac{\partial L}{\partial h_{t-1}} \cdot \frac{\partial h_{t-1}}{\partial h_{t-2}} \times \ldots \frac{\partial h_{t-k-1}}{\partial h_{t-k}}}$$

Now since the value of each derivative lies in between (0 to 1), the final answer ≈ 0. ↳ (tan)

$\boxed{\therefore \nabla L \approx 0} \Longrightarrow$ Update ≈ 0

∴ This is called as vanishing gradients.

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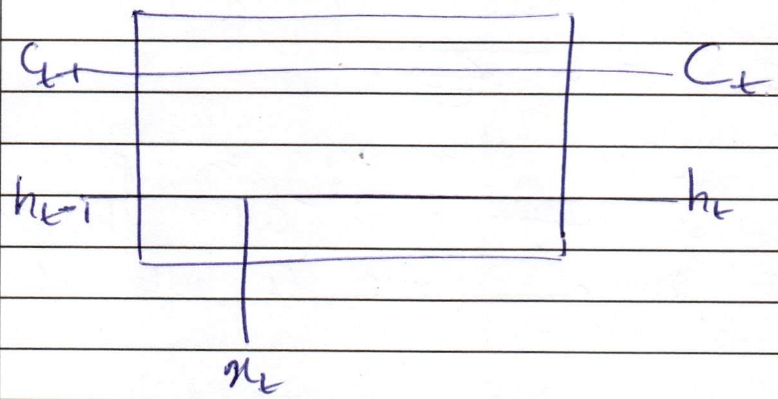
Q3) a) Contd Now, LSTM is long short term memory model which helps to solve this error using 2 memory states.



⇒ The C_t state is for long term memory. while h_t is for short term.

⇒ 3 gates are used ⇒ forget gate, input gate & o/p gate.

$$\boxed{C_t = f_t * C_{t-1} + i_t * \hat{C}_t}$$

Here, if $f_t = 0 \rightarrow$ it will forget the state
if $f_t = 1 \rightarrow$ it will pass it forward (h_{t-1})

Similarly if model wants to learn a new input it uses input gate.

if $i_t = 0 \rightarrow$ it doesn't take input
$i_t = 1 \rightarrow$ it takes candidate value.

~~$f_t = \sigma\, W[h_{t-1}$~~ $\boxed{f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)}$

$$\boxed{i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)}$$

Now, these eq^n are similar to RNN.

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	Q3) a) contd	suppose, this is the matrix that is to be multiplied.

$$\underbrace{\begin{bmatrix} 0 & 0 & \text{---} \\ 1 & 1 & \text{---} \\ \text{---} & & \\ \text{---} & & \end{bmatrix}}_{f_t} \quad \underbrace{\begin{bmatrix} x & y & \text{---} \\ z & a & \text{---} \\ \text{---} & & \\ \text{---} & & \end{bmatrix}}_{\hat{C}_t}$$

Since in LSTM elementary multiplication happens, 0s ~~allow~~ don't allow the data to pass through, while 1s allow the context to retain.

~~$\Rightarrow C_t = f_t * C_{t-1} + 0$~~

$$C_t = 0 * (h_{t-1}) + i_t * (\hat{C}_t)$$

$$= \underbrace{\begin{bmatrix} 0 & 0 & \text{---} \\ 1 & 1 & \text{---} \\ \text{---} & & \\ \text{---} & & \end{bmatrix}}_{(i_t)} * \underbrace{\begin{bmatrix} x & y & \text{---} \\ z & a & \text{---} \\ \text{---} & & \\ \text{---} & & \end{bmatrix}}_{(\hat{C}_t)}$$

$$= \begin{bmatrix} 0 & 0 & \text{---} \\ z & a & \text{---} \\ \text{---} & & \\ \text{---} & & \end{bmatrix}$$

Thus context is preserved.

Q2) a) Computation complexity of standard 2D convolution involves ~~an~~ input image of dimensions '$n \times n$' & ~~fiter~~ filter ~~of size~~ '$f \times f$'.

[I/P matrix] * [filter matrix] → [O/P matrix]

$(n \times n)$ $(f \times f)$ $(n-f+1) \times (n-f+1)$

eg. If $n = 4$, $f = 2$ then after convolution operation the O/P size would be '3'.

Thus the O/P image size is $(n-f+1) \times (n-f+1)$ — (i)

~~This is also~~

Thus no. of multiplication in standard 2D convolution ?

One element → $f \times f$ multiplications.

$\boxed{\therefore \text{ Total multiplications} = (n-f+1) \times (n-f+1) \times f \times f}$

from (i)

— Now consider case of K filters (eg. RGB has 3 filters).

To convolute filter has to be same dimension like if I/P → $H \times W \times C$, filter needs to have → $f \times f \times C$.

$(H \times W \times C)$ $(f \times f \times C)$ ~~$H \times W$~~ $(H-f+1) \times (W-f+1) \times$

[I/P matrix] * [K filters] = [O/P image]

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	Q3) b) contd	O/p size = $(H-f+1) \times (W-f+1)$
		But 2 K. kernels are used so,
		O/p = $(H-f+1) \times W-f$

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